

Expo and Defense

Project: Helpful Classroom Sorter

1. Your mission

Your learning target

I can evaluate a saved candidate and a manual baseline fairly.

I can defend a conclusion using results, failures and limits.

I can explain the rules, privacy choices and human control in your project.

Remember before reading

Close your notes first. A rough first answer helps us choose the right starting point; it is not a score for your ability.

Without opening the old lesson, explain one key idea from Improve One Cause. Give an example and a mistake that the idea helps you avoid.

Without opening the old lesson, explain one key idea from Test Normal Cases. Give an example and a mistake that the idea helps you avoid.

Without opening the old lesson, explain one key idea from Choose Inputs and Outputs. Give an example and a mistake that the idea helps you avoid.

After trying, check the relevant lesson or ask your teacher. Change the part of your explanation that the evidence shows was wrong.

Understanding is shown by explanation, transfer, testing, and limits.



LEARNING TARGET Defend a sorter with 10/12 routine successes and 2/4 mixed-material successes. Report overall and separate results, show one failure and propose a justified next step. What claim would overstate the evidence?

Learner name:

Readiness for the connected explanation

You change a candidate after seeing a final test fail. Can the same case be an unseen final test of the new version? Explain.

This week's end goal

By the end, I can defend a project using measured results, limits and an honest baseline comparison.

2. Notice the evidence

PICTURE STORY

Look Closely Before You Explain

Pictures give clues; evidence tells us what the clues mean.



MYSTERY A team presents the Classroom Sorter, traces a new item live, shows test evidence, explains one failure, and states when a human must decide.

What do you notice?

Name three visible details without explaining them yet.

Circle the detail most connected to the system's decision.

Name one detail that may be decoration or distraction.

Make a first prediction and state the clue that supports it.

My observations, prediction, and evidence:

GOOD THINKING A prediction can change when stronger evidence appears.

A closer explanation

This week you will evaluate your activity adviser and explain what its evidence supports. A project with an honestly reported failure can show strong understanding. Save your candidate instructions, version label, input rules, development records and known problems before your teacher reveals the final requests and catalogue. Keep the instructions unchanged during this evaluation.

The agreed rule remains: validate whole available minutes from 0 to 30 and Quiet required Yes or No. No allows quiet or nonquiet activities. Start at the first row and suggest the first activity whose time is no more than available time AND whose quiet label satisfies the request. Explain the choice. If all rows fail, report no match. Invalid or missing input needs clarification before searching. Clear the previous output for each run.

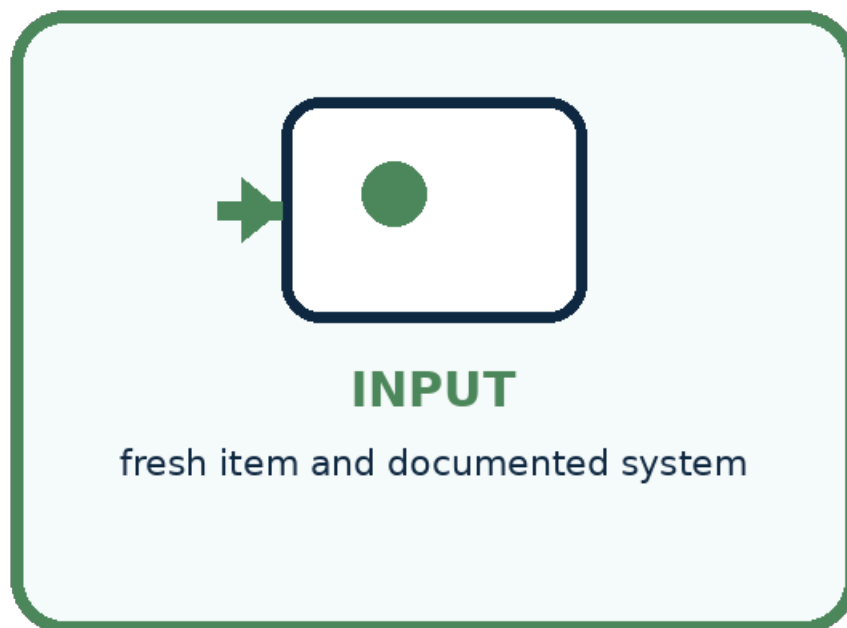
Your teacher will supply a fresh, labelled catalogue in the same format. Its values are test inputs, not permission to rewrite your instructions. All materials are assumed available. The manual baseline checks every row against the same requirements, then selects the first fit. Both methods advise only; neither books, publishes or chooses on a person's behalf.

3. See how it works

VISUAL MECHANISM

Input → Process → Output

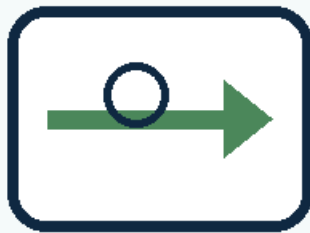
Follow the information instead of imagining magic inside the machine.





PROCESS

execute and explain each stage under questions



OUTPUT

result, evidence, limitation, and responsible next action

INPUT fresh item and documented system

PROCESS execute and explain each stage under questions

OUTPUT result, evidence, limitation, and responsible next action

CORE IDEA Understanding is shown by explanation, transfer, testing, and limits.

UNDER THE HOOD A defense evaluates transferable understanding, not presentation polish. Strong evidence includes system architecture, version history, planned tests, failures, controlled improvement, fairness/privacy boundaries, and independent response to a novel case.

Draw a fresh example with all three stages:

A closer explanation

From saved version to supported claim

Before the run · During the run · After the run

Before the run: Save instructions and label

During the run: Use the assigned data

After the run: Keep results and traces

Before the run: Keep final cases reserved

During the run: Run both methods fairly

After the run: Compare full passes

Before the run: State rules and limits

During the run: Record actual outputs

After the run: Explain failures and limits

Changing instructions during evaluation creates a different version. Keep evidence for the original version.

Use the same final requests, catalogue, requirements and full-pass rule for both methods. Record the candidate output and reason before looking at the expected answer. Record the baseline separately; do not copy the candidate result into it. Your teacher holds the answer key. A full pass needs both the required output and a valid reason, including why earlier rows fail when a later row is selected.

For each request record its ID, inputs, catalogue label, candidate label, candidate output and reason, baseline output and reason, and both results after checking. Keep a trace for any failure. Mark unrun cases explicitly. Do not silently correct an observation. If a person misreads an instruction, record that execution error separately from a defect in the written instruction.

Connect the picture and the explanation

Point to the input, the deciding step and the result in this week's visual. Explain the same step in ordinary words. A picture helps when you can say what each part represents.

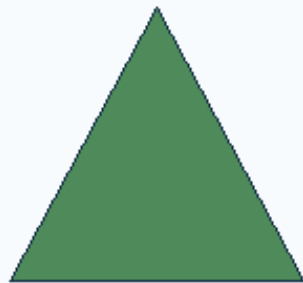
4. Compare the cases

Four Cases, Four Clues

Use the visual cue, then read the evidence carefully.



Polished poster



Fresh trace



Failure log



Human gate

CASE 1 — Polished poster Clue: communication aid. Meaning: not proof alone.

CASE 2 — Fresh trace Clue: unrehearsed application. Meaning: transfer evidence.

CASE 3 — Failure log Clue: honest testing record. Meaning: engineering evidence.

CASE 4 — Human gate Clue: responsible boundary. Meaning: use condition.

BOUNDARY Confidence, decoration, or memorized vocabulary cannot replace a fresh trace and evidence-based answers.

Which two pictures are easiest to confuse? What evidence separates them?

Change one thing in the pictured case

Change: Someone adds the two unrun cases to the denominator as failures.

Prediction: That reports a different measure, 8/12 completed successes out of planned cases.

Reason: It cannot be called the original attempted-case pass rate; distinguish completion from evaluated correctness.

5. Words that unlock the idea

VOCABULARY

Words You Can See and Use

The badge is a memory cue; the definition does the real thinking work.





defense



transfer



Word · Meaning · My example

Word: demonstration

Meaning: live showing of system behavior

My example: _____

Word: defense

Meaning: evidence-based explanation under questions

My example: _____

Word: transfer

Meaning: applying understanding to a fresh case

My example: _____

Word: limitation

Meaning: condition where performance or use is restricted

My example: _____

Misconception detective

CLAIM TO REPAIR A successful demo proves every case will work.

Use one vocabulary word and one piece of evidence to repair it:

WORD POWER A strong definition helps you decide something in a new case.

Words used in the closer explanation

Word · Meaning

Word: Final evaluation

Meaning: Testing a saved candidate under stated conditions after development

Word: Baseline

Meaning: A comparison method tested under the same conditions

Word: Evidence

Meaning: Recorded observations that support or challenge a claim

Word: Defence

Meaning: An explanation of a decision using evidence and reasoning

6. Follow a worked example

WORKED EXAMPLE

Watch the Reasoning, Not Just the Answer

Every step connects a claim to observable evidence.



CASE A team presents the Classroom Sorter, traces a new item live, shows test evidence, explains one failure, and states when a human must decide.

STEP 1 — OBSERVE fresh item and documented system

STEP 2 — TRACE execute and explain each stage under questions

STEP 3 — CONCLUDE result, evidence, limitation, and responsible next action

MODEL REASONING A complete defense traces input, process, and output; distinguishes rules from learned parts; presents normal and edge tests; explains one controlled improvement; states privacy, fairness, uncertainty, and human-review boundaries; and handles a fresh case independently.

DO NOT OVERCLAIM Confidence, decoration, or memorized vocabulary cannot replace a fresh trace and evidence-based answers.

What new evidence could change this answer?

A closer explanation

Here is a fictional example from a different project, not your class results. A candidate has 6 full passes from 8 attempted cases; the baseline has 8 from the same 8. Two more planned cases were not run. The candidate passed 75 percent of attempted cases; the baseline passed 100 percent. Neither has evidence for the two unrun cases. The candidate did not outperform the baseline on this measure.

Evidence · Supported statement · Claim to avoid

Evidence: 6 of 8 full passes

Supported statement: Six attempted cases met the full rule

Claim to avoid: It works for every request

Evidence: Both methods pass the same cases

Supported statement: They tie on this measured result

Claim to avoid: Automation must be better

Evidence: A time was not measured

Supported statement: Speed is unknown

Claim to avoid: The candidate is faster

Evidence: Only fictional catalogue data

Supported statement: Results concern these test conditions

Claim to avoid: People will enjoy every suggestion

Explain each failure and its consequence instead of hiding it inside a total. Even perfect results on a small reserved set do not establish reliability in every situation. Our paper adviser applies human-written rules; it has not learned a model from training examples. It checks compatibility, not enjoyment, popularity or real material availability.

In the fictional example, state the candidate and baseline results, explain how the two unrun cases should be reported, and give one useful next action. Why is “the adviser is ready for every classroom” unsupported?

Learn from the worked case

Cover the result before checking it. Say what is given, choose the procedure, and follow one step at a time. Then uncover the result and explain your first mismatch. Ask why a step is allowed, not only what number it produces.

A pictorial worked case: Expo and Defense

A fictional paper sorter team prepares an exhibition evidence board.

The facts we will use

Both prototype and manual baseline attempt the same 10 reserved cases. Prototype passes 8; baseline passes 9.

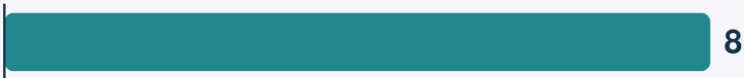
Two additional planned cases are unrun. No timing or pupil-satisfaction measures were collected.

AI ACADEMY / AI-1/36

Shared reserved-case comparison

passes out of 10 attempted

Prototype



Manual baseline



Bar length shows the supplied value; the marked baseline is zero.

Expo and Defense

Start with the supplied facts

Follow each arrow. Ask which fact allows the next step.



1. Describe contract

Show the task, inputs, rule and reserved case selection.



2. Show results

Prototype $8/10=80\%$; baseline $9/10=90\%$ on attempted cases.



3. Keep incomplete work

Two unrun cases remain visible, without pass/fail labels.



4. Defend conclusion

Prototype did not beat baseline on this measured outcome; explain its failures.

CHANGE ONE THING

Someone adds the two unrun cases to the denominator as failures.

Explain the changed result using the matching lesson.

Read the picture one step at a time

Step · What happens and why

Step: 1. Describe contract

What happens and why: Show the task, inputs, rule and reserved case selection.

Step: 2. Show results

What happens and why: Prototype $8/10=80\%$; baseline $9/10=90\%$ on attempted cases.

Step: 3. Keep incomplete work

What happens and why: Two unrun cases remain visible, without pass/fail labels.

Step: 4. Defend conclusion

What happens and why: Prototype did not beat baseline on this measured outcome; explain its failures.

Match each pictured step to the supplied facts. This is a worked teaching case, not an independent assessment.

Why this result follows

An exhibition should make the evidence inspectable. A polished demonstration is not a substitute for the reserved-case record. The shared cases allow a direct descriptive comparison of pass rates; missing timing data prevents a claim that the prototype is faster. Unrun cases must not be treated as successes.

What this example does not establish

This fictional evaluation does not demonstrate classroom learning impact or performance on every future request.

Read the labelled picture: Write one defensible success statement and one claim the evidence cannot support.

Change one thing: Someone adds the two unrun cases to the denominator as failures. Predict the result before checking the explanation. What remains the same?

7. Find and repair a mistake

COMPARE

Same Result, Different Reason

A comparison helps reveal the true concept boundary.

Case · Evidence · What it shows

Case: Polished poster

Evidence: communication aid

What it shows: not proof alone

Case: Fresh trace

Evidence: unrehearsed application

What it shows: transfer evidence

Case: Failure log

Evidence: honest testing record

What it shows: engineering evidence

Case: Human gate

Evidence: responsible boundary

What it shows: use condition

Repair two tempting claims

CLAIM A A successful demo proves every case will work.

Repair:

CLAIM B Admitting a limitation lowers the quality of the project.

Repair:

STRONG ANSWER FRAME I claim _____ because the evidence shows _____. The process is _____. One limit is _____.

A closer explanation

Archive the evaluated candidate and its original results. If you repair it after evaluation, give it a new version label and use revealed cases as development checks. Do not relabel them as unseen. Ask your teacher for fresh reserved cases only after the revision is fixed. Do not erase the first result.

Write one conclusion limited to the actual conditions tested and one specific next step. It may be better to keep the manual baseline. Grade 7 will turn carefully stated procedures into programs and examine data more closely. Carry forward the habits of validating inputs, recording evidence and keeping people in control.

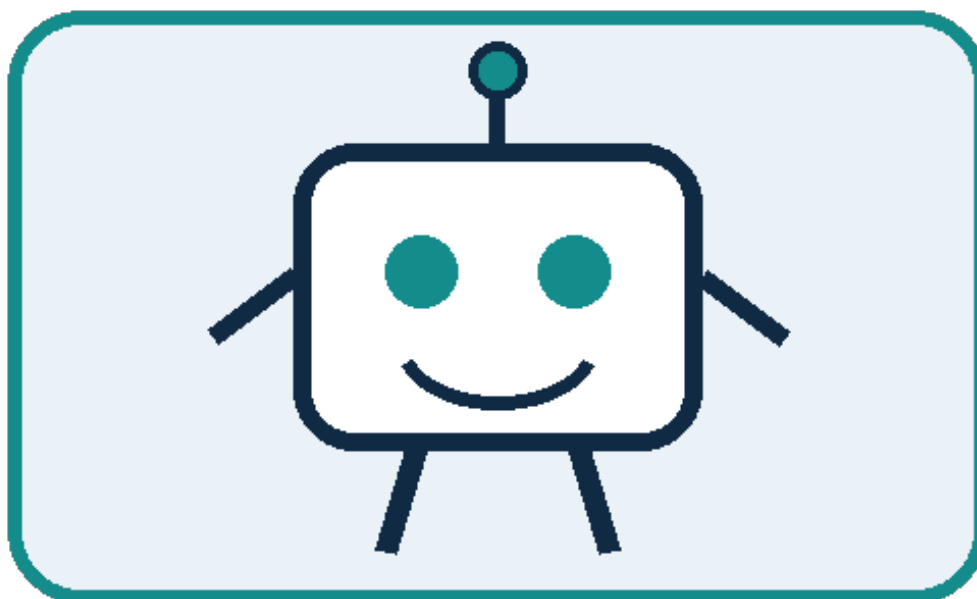
Name one observation from your own evaluation, one limit on your conclusion and one next action. Explain why your adviser is a rule-based program design rather than a trained model, and why passing tests does not give it permission to book an activity.

8. Practise together

GUIDED PRACTICE

Try It, Check It, Explain It

Use support now so you can succeed independently later.



Explain how the picture story demonstrates this idea: Understanding is shown by explanation, transfer, testing, and limits.

Use a fresh example to explain demonstration.

Trace a new input → process → output example.

Find and repair the error in: Admitting a limitation lowers the quality of the project.

Invent one edge case and name the safe response.

PARTNER CHECK Point to the evidence. Do not accept an answer that only repeats the vocabulary word.

A closer explanation

Prepare a short explanation using your saved instructions, one complete trace, the final comparison, and a limitation. State the problem and whose choice the adviser supports. Explain validation, both matching conditions, first-fit order, reset and stopping. Show the exact instruction behind your trace. Describe any repair with its old evidence, or say honestly that no repair was needed.

Use only fictional requests and the two necessary input fields. Do not collect names, diagnoses or reasons a pupil prefers quiet. A quiet request is enough. A person may decline the suggestion; a teacher checks real materials and local rules before an activity. A successful output grants no permission to book or publish.

For a two-minute partner presentation, use this order: problem and rule; trace; candidate versus baseline results; failure or limitation; next action. Partners swap speaker and listener roles. Each person must explain one trace independently. The listener asks which recorded observation supports the main claim; attractive slides cannot substitute for that observation.

Practise with less help: Keep the final evaluation record

Use workbook W2 for the connected example “Evaluating and explaining your project”. First fill the input and rule boxes. Try the middle steps before asking for a hint. After a hint, try the next step yourself.

Plan · Do · Check

Plan: What is given? Which rule or procedure applies?

Do: Record each intermediate step.

Check: Does the result follow? What remains unknown?

Explain your trace to a partner. Your partner points to one step to check. Swap roles on the next case. Each person writes their own explanation.

Check your reasoning

Use the worked case for Evaluating and explaining your project. Proposed explanation: “A polished presentation proves the system works”. Decide whether this explanation is supported. Point to the step or supplied fact that decides; explain your repair if needed.

Answer on your own before discussion. If you are unsure, say which step you need help with.

Explain the picture

Write one defensible success statement and one claim the evidence cannot support.

My explanation using a pictured step and the given facts

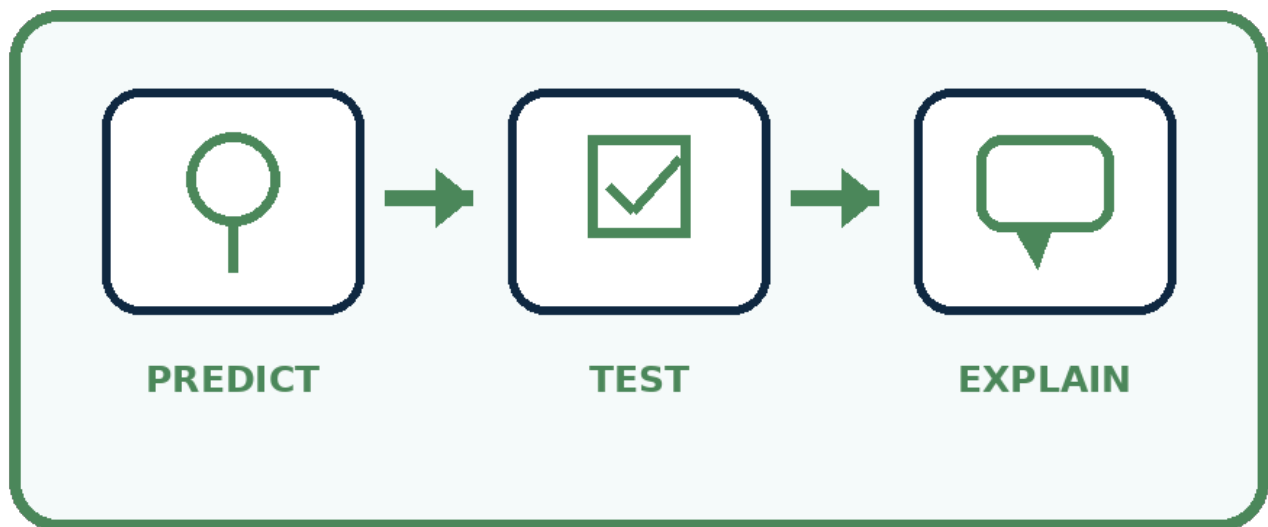
This is supported practice. Keep the separately named independent case for an attempt before feedback.

9. Run the investigation

EXPLORER LAB

Responsible Classifier Expo

Predict first, test honestly, and explain what the evidence means.



State your prediction before the result is known.

Complete the task: Demo the system, solve a fresh case, and answer why it may fail.

Keep a normal case and an edge case clearly separated.

Record what happened—even when it does not match your prediction.

Explain the result, one limitation, and the next useful test.

Case · Prediction · Observed result · What it means

Case: Normal

Prediction: _____

Observed result: _____

What it means: _____

Case: Edge

Prediction: _____

Observed result: _____

What it means: _____

INDEPENDENCE After support is removed, repeat the proof with a fresh case.

Plan → try → check

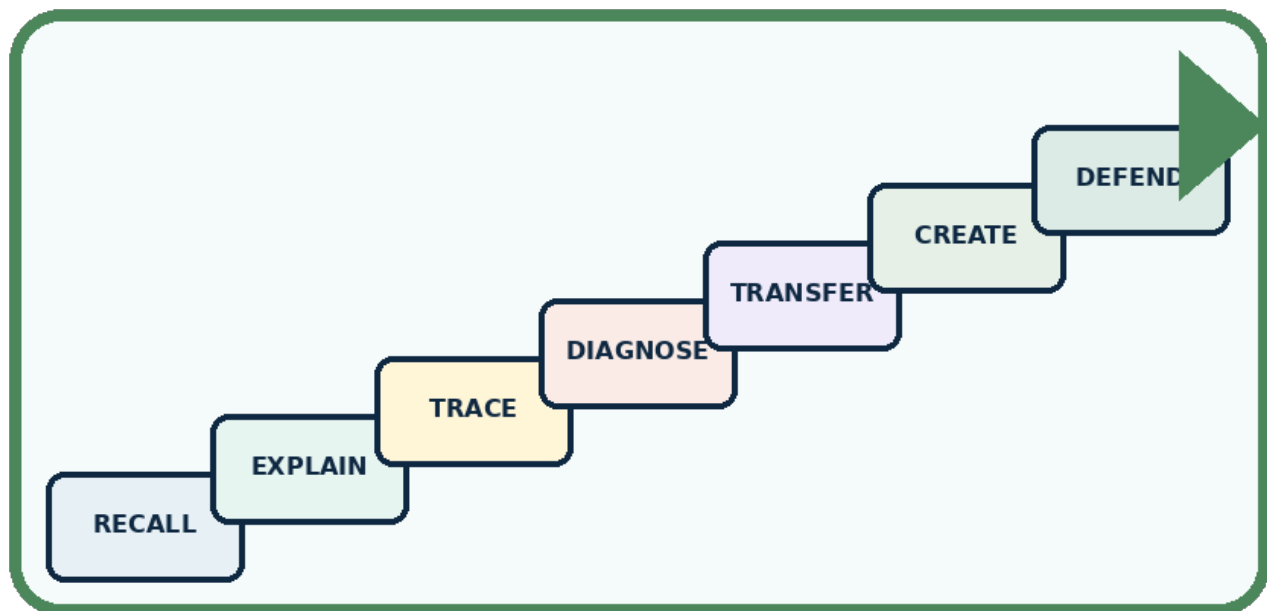
Before trying: what do I expect, and why? While trying: which step could first go wrong? Afterwards: what did I actually observe, and what should change? Keep “not run” if you only traced the procedure.

10. Build your understanding

PRACTICE LADDER

Climb from Recall to Defense

Each step asks for a stronger kind of understanding.



RECALL Define demonstration.

EXPLAIN Understanding is shown by explanation, transfer, testing, and limits.

TRACE Show the input, process, and output.

DIAGNOSE Repair: A successful demo proves every case will work.

TRANSFER Apply the idea to a new home or classroom case.

CREATE Design one revealing test.

DEFEND Give a claim, evidence, mechanism, limit, and safe action.

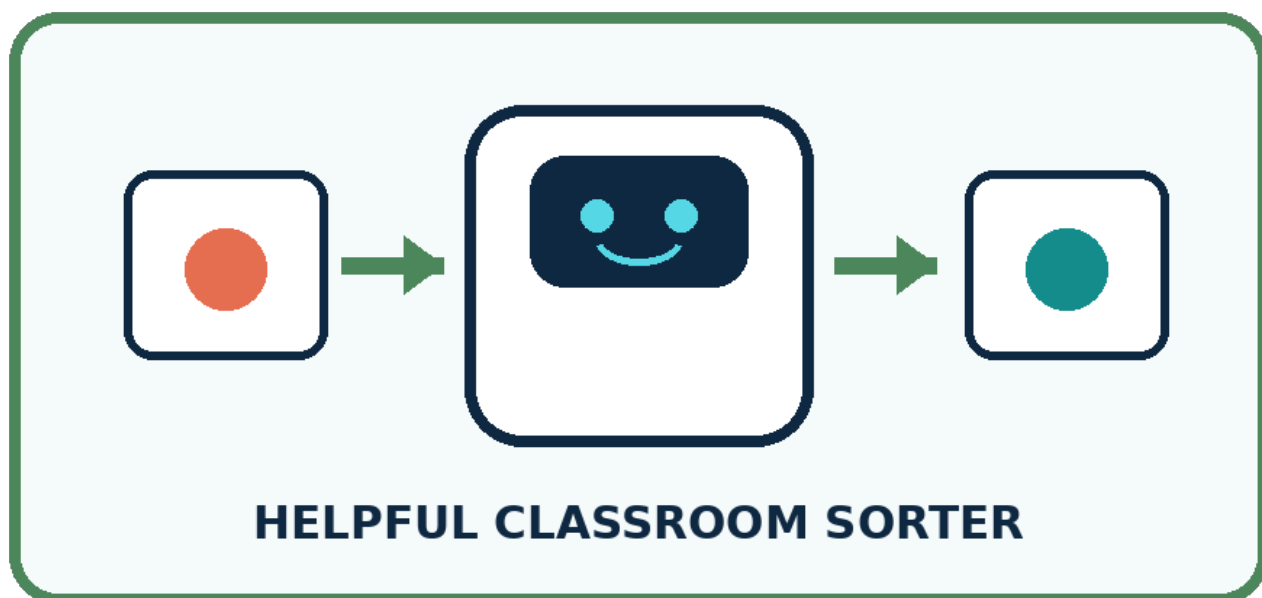
RETRY RULE A coached answer becomes mastery only after you solve a different case independently.

11. Connect your project

PROJECT

Helpful Classroom Sorter

Build one coherent system across the year, one evidence-based decision at a time.



THIS WEEK'S CHECKPOINT Present the final Classroom Sorter portfolio and complete the AI-1 certification defense.

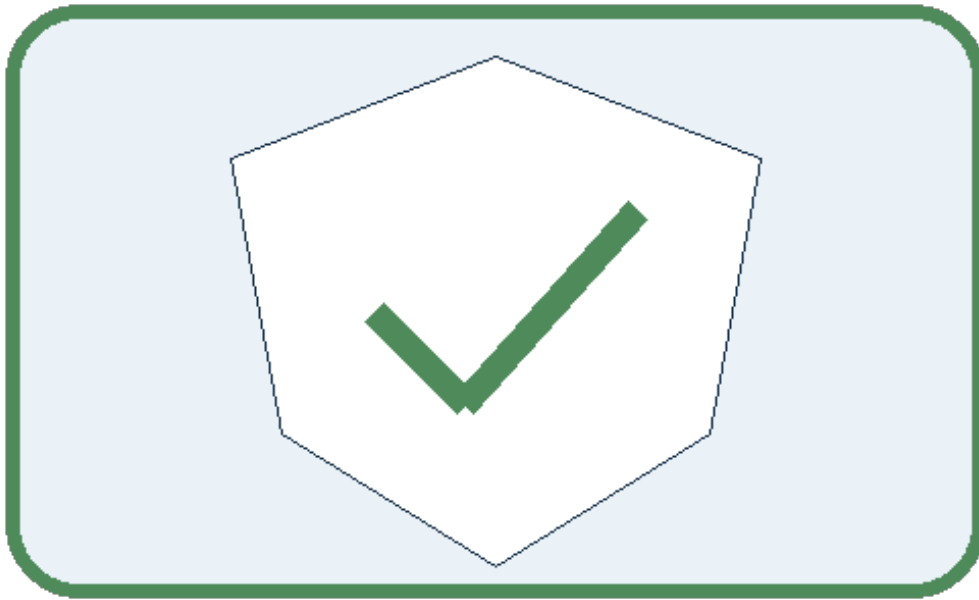
What real classroom need does this serve?

What information is necessary—and what should stay private?

What happens inside the process?

What normal and edge cases will you test?

When should it say not sure or ask a person?



RESPONSIBLE DESIGN Confidence, decoration, or memorized vocabulary cannot replace a fresh trace and evidence-based answers.

My project decision:

Evidence for your project

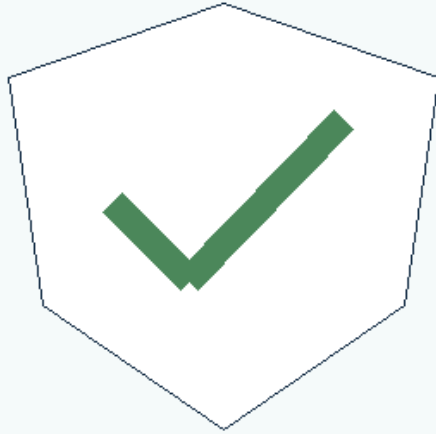
Present the Helpful Classroom Sorter's versions, test records, baseline, failures and next investigation as an evidence portfolio.

12. Show what you know

MASTERY

Fresh-Case Independent Proof

Complete this after teaching and guided practice have ended.



INDEPENDENT PROOF

REQUIRED PROOF Demo the system, solve a fresh case, explain test evidence, name limits, and answer why the system may fail.

- ☐ Name or draw the fresh case.
- ☐ Trace input, process, and output.
- ☐ State the conclusion and strongest evidence.
- ☐ Name one limit, failure, missing clue, or fairness/privacy concern.
- ☐ Explain one safe next action or improvement.

Independent evidence:

SELF-CHECK A complete defense traces input, process, and output; distinguishes rules from learned parts; presents normal and edge tests; explains one controlled improvement; states privacy, fairness, uncertainty, and human-review boundaries; and handles a fresh case independently.

My level: ☐ Not yet ☐ With one hint ☐ Independently ☐ I can teach it

NEXT BRIDGE Carry today's evidence habit into: the next Academy level.

13. Study a complete case

Week 36 Worked case

Use the supplied details to practise the weekly concept before attempting the independent question. This

case extends the illustrated lesson.

The situation

The expo poster says always correct after six classroom passes.

The question

Write a defensible claim.

Worked explanation

Report six specified passes with conditions and limits. Show an input, decision trace, output and test record, including an uncertain case.

Explain it in your own words

Which detail supports the answer, and why does it matter?

Write or draw your reasoning and evidence.

Trace the supplied case visually

Defend a claim with a complete trace

Evidence or stage · Value or result

Evidence or stage: Input

Value or result: Show a specified test object

Evidence or stage: Decision

Value or result: Explain its path

Evidence or stage: Output

Value or result: Show the tray or review result

Evidence or stage: Evidence

Value or result: Report six classroom passes and limits

The expo poster says always correct after six classroom passes.

Write a defensible claim.

Report six specified passes with conditions and limits. Show an input, decision trace, output and test record, including an uncertain case.

14. Try a new case independently

Independent case: Judge a different project independently

A fictional library adviser and its manual baseline face the same 6 reserved cases. The candidate has 4 full passes; the baseline has 5. One additional planned case is unrun by both. No speed data was collected. The team says “ours is faster and more reliable everywhere” and repairs it after seeing the failures. Explain what the counts support, two unsupported claims and how to label a rerun of the revealed cases.

My independent explanation, trace and limits

Attempt this case before feedback. Record any help. Earlier examples below are further practice; a case already practised with feedback cannot establish fresh transfer.

Week 36 Independent application

Close the worked explanation. Apply the idea to this different question without copying.

At the expo, a team says our sorter never fails after testing four cards. Replace the claim with an evidence-based statement and name a next test.

My answer, with a trace, calculation or supporting evidence

Write or draw your reasoning and evidence.

Create and test

Change one relevant detail. Predict the result before testing. Include a boundary or missing-information case when it helps reveal a limit.

My changed input and expected result

Write or draw your reasoning and evidence.

Connect to your project

Present the final Classroom Sorter portfolio and complete the AI-1 certification defense.

The evidence I will keep

Write or draw your reasoning and evidence.

Your teacher has the independent answer and scoring criteria. Complete the matching workbook week to

keep your practice evidence together.

Transfer practice from the original programme

Defend a sorter with 10/12 routine successes and 2/4 mixed-material successes. Report overall and separate results, show one failure and propose a justified next step. What claim would overstate the evidence?

Explain the mechanism, evidence and one limit.

Use feedback to improve

Attempt the independent case before hints. Record whether you worked alone, used a hint or followed a model. After feedback, fix the first unsupported step and explain why it changed. A later check uses a changed case, not a copied answer.

15. Your lesson route

Start with this question: A fictional paper sorter team prepares an exhibition evidence board.

By the end, I can · How I will show it

By the end, I can: Evaluate a saved candidate and a manual baseline fairly

How I will show it: Explain the deciding step, show a trace or test, and state one limit.

By the end, I can: Defend a conclusion using results, failures and limits

How I will show it: Explain the deciding step, show a trace or test, and state one limit.

By the end, I can: Explain the rules, privacy choices and human control in your project

How I will show it: Explain the deciding step, show a trace or test, and state one limit.

Remember → watch and explain → try with help → investigate → try independently → keep project evidence. Your teacher may spread this lesson across several classes.

Keep your first prediction. Compare it with what the evidence shows and explain any change.

16. Finish and return

Close your notes. Explain this goal in your own words: Evaluate a saved candidate and a manual baseline fairly

My deciding step, evidence and one remaining question

If you are unsure, return to the worked picture and explain one arrow. If you are ready, change an input or assumption and justify your prediction.

Save your workbook evidence. At the next review, try a different case before opening your old answer.

17. Supported practice, repair and portfolio

Use this supported practice once, in the browser or on paper. Keep the reserved independent assessment for a separate first attempt before teacher feedback.

A fictional paper sorter team prepares an exhibition evidence board.

Practice 1: explain the deciding evidence

Which labelled step matches this detail? Prototype did not beat baseline on this measured outcome; explain its failures.

1. Keep incomplete work
2. Describe contract
3. Show results
4. Defend conclusion

My selected conclusion, deciding fact and explanation

Hint if needed: Read the steps in the pictorial case. Match the supplied detail to the stage that performs or records it.

After your attempt, compare with the supported explanation: Defend conclusion Prototype did not beat baseline on this measured outcome; explain its failures. An exhibition should make the evidence inspectable. A polished demonstration is not a substitute for the reserved-case record. The shared cases allow a direct descriptive comparison of pass rates; missing timing data prevents a claim that the prototype is faster. Unrun cases must not be treated as successes.

Practice 2: explain the deciding evidence

Changed-case practice: Someone adds the two unrun cases to the denominator as failures. Which conclusion is supported by the supplied facts?

1. We can decide the result without examining the changed input or the procedure.
2. This one example guarantees correct results for every future input.
3. That reports a different measure, 8/12 completed successes out of planned cases.

My selected conclusion, deciding fact and explanation

Hint if needed: Use the changed condition and the deciding step in the pictorial trace. The answer must say what follows from those facts.

After your attempt, compare with the supported explanation: That reports a different measure, 8/12 completed successes out of planned cases. It cannot be called the original attempted-case pass rate; distinguish completion from evaluated correctness.

Repair a missing building block

Review Improve One Cause (ai-1/35). Goal: Repair a located defect without changing the agreed

requirements. Short practice: Explain why “our repaired version now picks something” is not enough to show success at 6.. Return to this week and explain what you changed.

Review Test Normal Cases (ai-1/33). Goal: Plan routine cases with expected results derived from the requirements. Short practice: Why is none correct for 8 but incorrect for 6? Use both capacity cards.. Return to this week and explain what you changed.

Review Choose Inputs and Outputs (ai-1/30). Goal: Turn the project outcome into precise input and output requirements. Short practice: Trace requests 10 and blank. Why should they not receive the same explanation?. Return to this week and explain what you changed.

Keep your completed work

Use Download completed work in the hosted lesson to export your answers, reflections, practice feedback and code-run evidence. Open the HTML copy to read or print it. Drafts stay in this browser when storage is available; clear drafts on a shared device.